

Hisham Bhatti

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EDUCATION

University of Washington

M.S. Computer Science (GPA: 4.0)

Seattle, WA

Sept. 2025 – June 2027

University of Washington

B.S. Computer Science, B.S. Mathematics (GPA: 3.97)

Seattle, WA

Sept. 2021 – June 2025

EXPERIENCE

Software Engineer Intern

Google

June 2026 – Sept. 2026

San Diego, CA

- Developing a framework to analyze Android battery Trusted Firmware-A Tests latency data

AI Engineer Intern

Revefi

Apr. 2025 – June 2025

Redmond, WA

- Implemented data quality monitors across four major data sources (Snowflake, Databricks, Redshift, BigQuery), enabling companies to track invalid entries with SQL-based metrics
- Integrated these checks into an AI-powered recommendation system, reducing manual setup time and contributing to a platform-wide **10×** improvement in operational efficiency and **60%** reduction in data spend

Undergraduate Researcher: Topological Algorithms

U.S. National Science Foundation

June 2024 – Sept. 2025

Chico, CA

- Developed an algorithm in SageMath to construct Tait graphs for **27,000+** knots, contributing new functionality to the open-source SageMath library used by **300,000+** researchers

PROJECTS

Mini Google | C/C++, POSIX, Multithreading, Web Development

Jan. 2024 – Mar. 2024

- Built a multithreaded file search engine in C/C++ with a doubly-linked list and hash table for document indexing
- Developed a POSIX-compliant web server handling **50+** concurrent users, with HTML rendering and input sanitization for secure file access

PINO Robustness | Python, PyTorch, Scientific ML | GitHub

Mar. 2026 – June 2026

- Evaluated robustness of Physics-Informed Neural Operators (PINO) across **32×** zero-shot super-resolution, demonstrating stable PDE solution accuracy far beyond training resolution
- Conducted resolution ablations across training grids from **4–128** points, identifying failure regimes where physics and data constraints diverge
- Integrated sparse real-world AAPL options data into a physics-informed learning pipeline, achieving solution quality comparable to analytic supervision

CBF-RL for Dynamic, Multi-Agent Navigation | PyTorch, PPO, Robotics | GitHub

Apr. 2026 – June 2026

- Extended Control Barrier Function Reinforcement Learning (CBF-RL) from single-agent navigation to multi-agent double-integrator dynamics with stochastic control noise
- Built a parallel PPO training loop with **4,096** simultaneous environments, enabling large-scale safety evaluation
- Executed a **68-experiment** ablation study across safety filters, reward shaping, and agent-obstacle densities, achieving up to **96.7%** navigation success

PUBLICATIONS

On the Structure of Bad Science Matrices. *Involve*, 2024. [Paper](#)

A Theoretical Assessment of Nanopore Protein Fingerprinting. *Biophysical Journal*, 2024. [Paper](#)

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, MATLAB, Bash

AI/ML: PyTorch, TensorFlow, Claude Code, Cursor

Frameworks: React, Node.js, Flask, JUnit, Spring Boot

Developer Tools: Git, Docker, Google Cloud Platform, Azure, AWS, Linux/Unix